TP_modul_7

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NIM + 108092500012, digit terakhirnya adalah 2. 2 MOD 4 = 2, output "PERCAYADIRI" dan "DIRI".

FILE CODE

1. stack.h

stack.h | stack.cpp | main.cpp

```
#ifndef STACK_H
#define STACK_H

typedef char infotype;

struct stack {
    infotype info[15];
    int Top;
};

#define Top(S) (S).Top
#define info(S) (S).info

void createStack(stack &S);
bool isEmpty(stack S);
bool isFull(stack S);
void push(stack &S, infotype x);
infotype pop(stack &S);
void printInfo(stack S);

#endif
```

2. stack.cpp

stack.h | stack.cpp | main.cpp

```cpp
#include "stack.h"
#include <iostream>

using namespace std;

void createStack(stack &S) {
    Top(S) = 0;
}

bool isEmpty(stack S) {
    if (Top(S) == 0) {
        return true;
    } else {
        return false;
    }
}

bool isFull(stack S) {
    if (Top(S) == 15) {
        return true;
    } else {
        return false;
    }
}

void push(stack &S, infotype x) {
    if (isFull(S) == false) {
        Top(S) = Top(S) + 1;
        info(S)[Top(S) - 1] = x;
    }
}

infotype pop(stack &S) {
    infotype x = info(S)[Top(S) - 1];
    Top(S) = Top(S) - 1;
    return x;
}

void printInfo(stack S) {
    for (int i = Top(S); i >= 1; i--) {
        cout << info(S)[i - 1];
    }
    cout << '\n';
}
```

3. main.cpp

stack.h | stack.cpp | main.cpp

```cpp
#include "stack.h"
#include <iostream>

using namespace std;

int main() {
    stack S;
    createStack(S);

    push(S, 'I');
    push(S, 'R');
    push(S, 'I');
    push(S, 'D');
    push(S, 'A');
    push(S, 'Y');
    push(S, 'A');
    push(S, 'C');
    push(S, 'R');
    push(S, 'E');
    push(S, 'P');

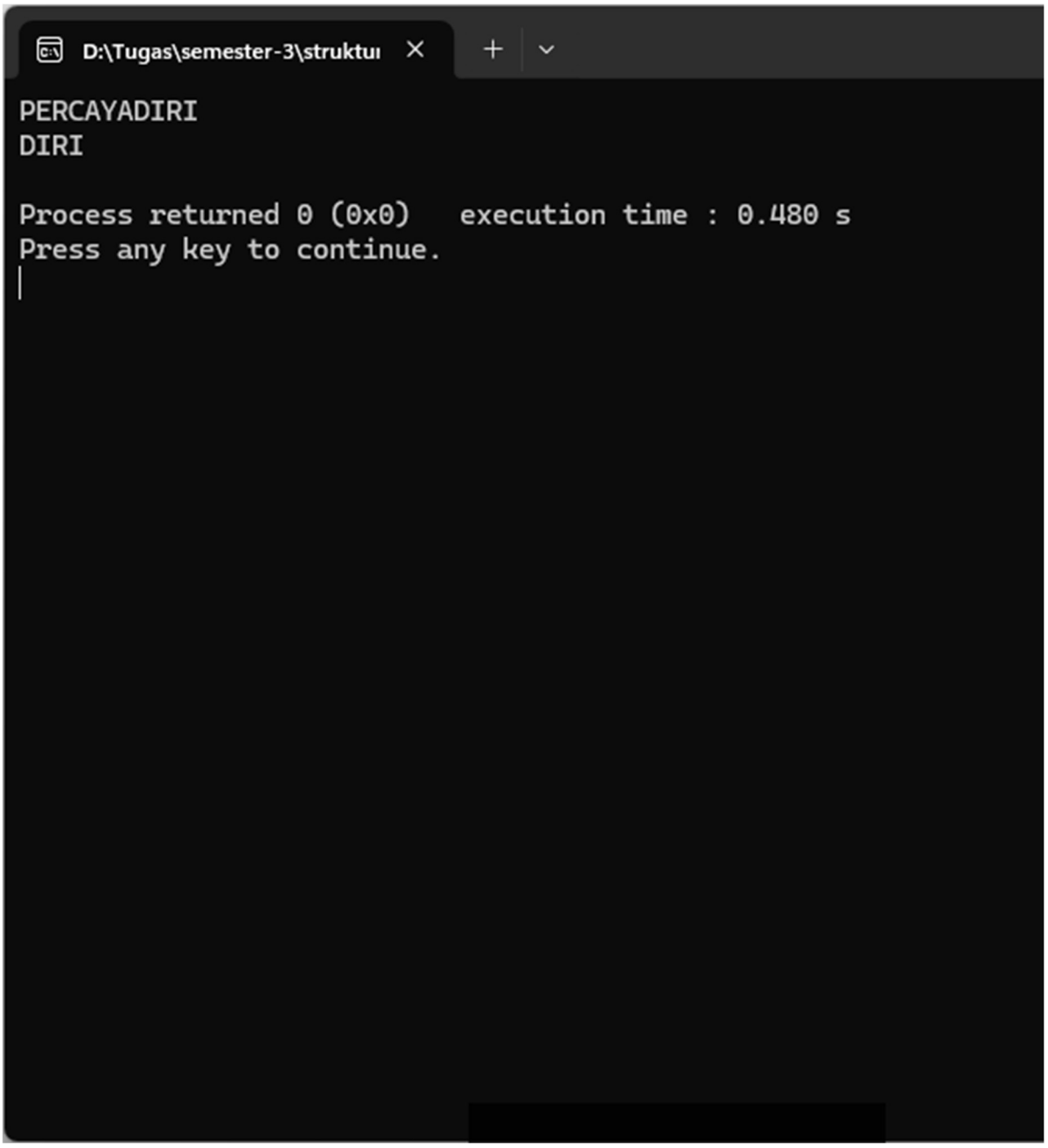
    printInfo(S);

    for (int i = 0; i < 7; i++) {
        pop(S);
    }

    printInfo(S);

    return 0;
}
```

BUILD & RUN


D:\Tugas\semester-3\struktur
PERCAYADIRI
DIRI
Process returned 0 (0x0)   execution time : 0.480 s
Press any key to continue.